CENG0065 Applications of Digital Materials Manufacturing

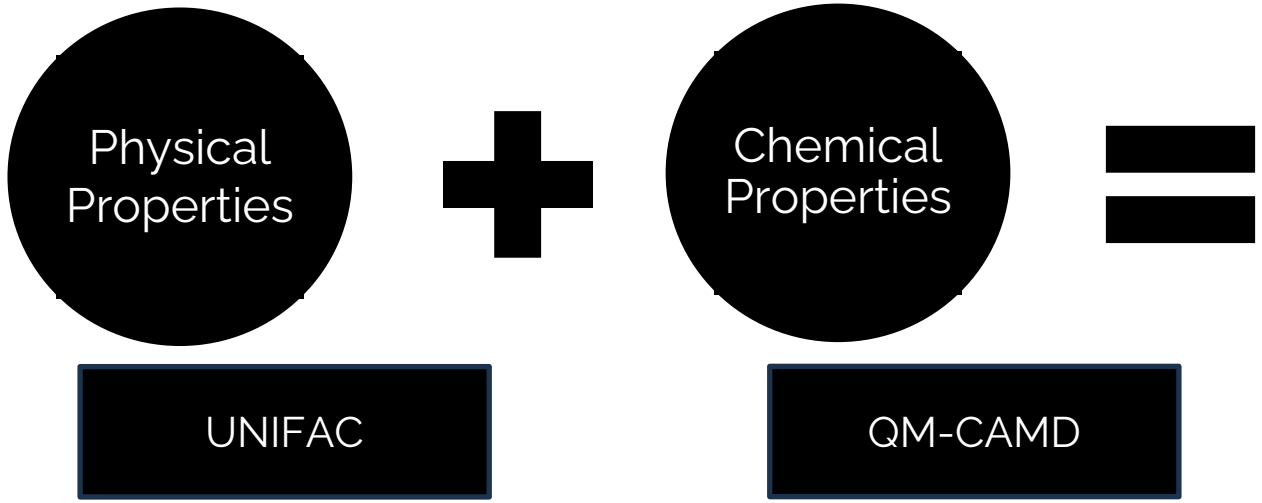


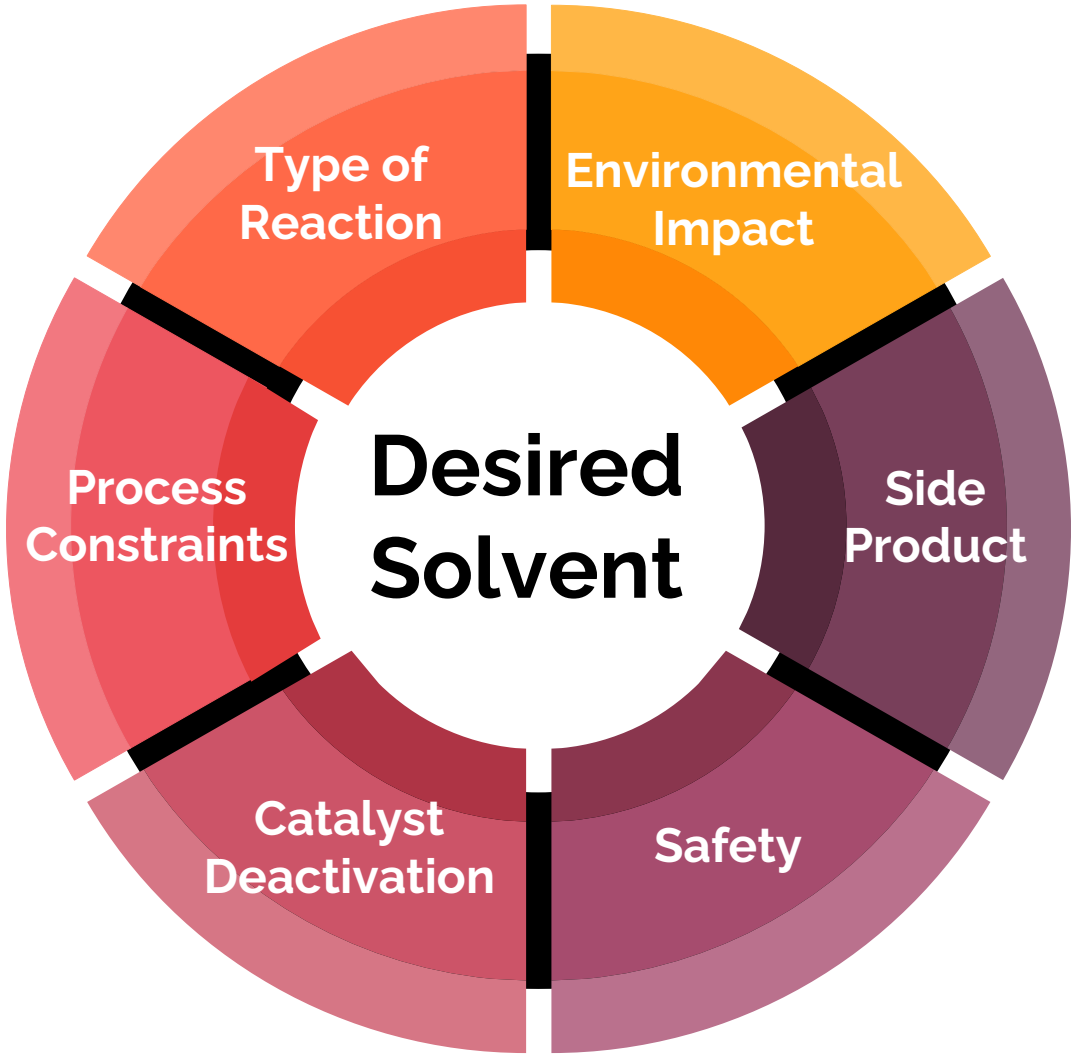
Computer-aided Molecular Design of Solvents for Accelerated Reaction Kinetics

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Introduction & Motivation





UNIFAC = Universal quasichemical functional-group activity coefficients

QM-CAMD = Quantum mechanical computer-aided molecular design

Introduction & Motivation

Conventional QM-CAMD & Other Methods	V.S	Proposed QM-CAMD
Required Large Data		Required Less Data
Neglect Solvent Effect on Reaction Kinetic		Include Reaction Rate Constant
Large Computational Cost		Low Computational Cost

Methodology

❑ Proposed algorithm and key concepts

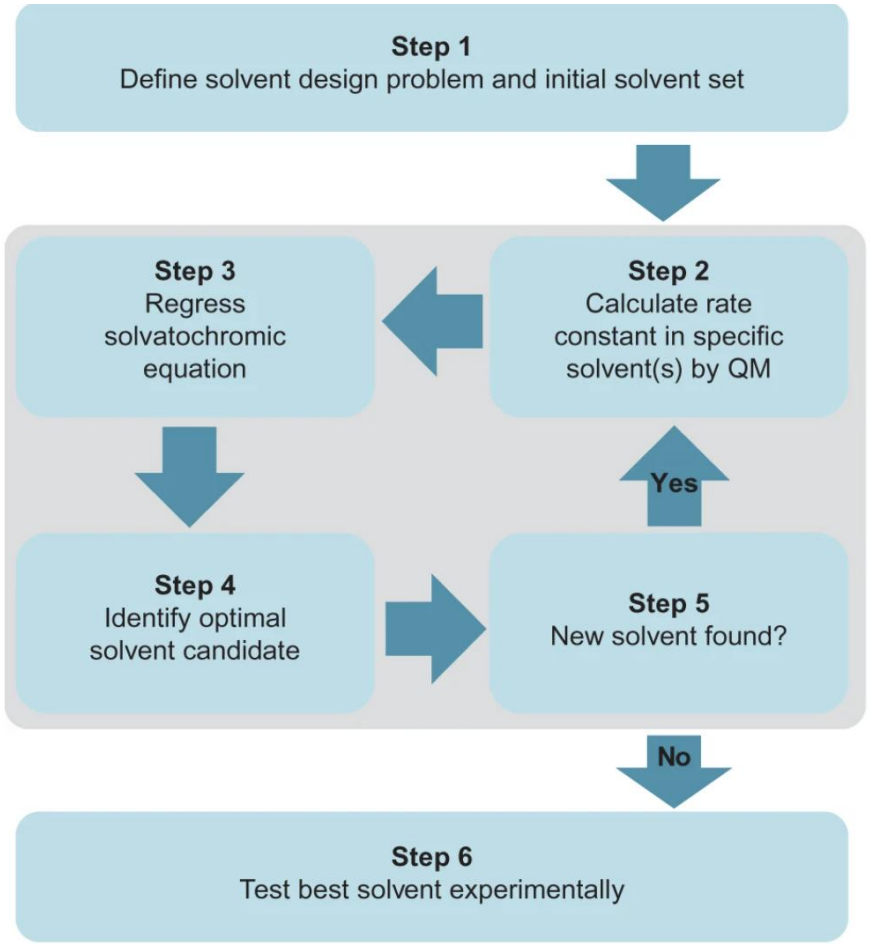


Figure 1: CAMD algorithm presented in the paper [1]

Concept related to the Module

- CAMD
- Group-Contribution (GC)
- Reaction Kinetic
- Mixed Integer Linear Optimization

New Concept

- SMD + QM density functional theory (DFT)

Algorithm

- **Defined Objective**: maximize the reaction rate constant k at 298K [1]
- **Set Constraints**: boiling points and melting points [1]
- **Give Initial set of six solvents** with varying polarity and functional groups [1]

Table 1 | Key properties for the six initial solvents and the best solvent identified, nitromethane.

Solvent	Dielectric constant*	Rate constant $k^{QM\dagger}$ ($dm^3\ mol^{-1}\ s^{-1}$)
Toluene	2.38	1.61×10^{-6}
Chlorobenzene	5.70	3.94×10^{-5}
Ethyl acetate	5.99	4.88×10^{-5}
THF	7.43	8.70×10^{-5}
Acetone	20.49	2.86×10^{-4}
Acetonitrile	35.69	4.70×10^{-4}
Nitromethane	36.56	6.09×10^{-4}

*Experimental dielectric constant at 298 K (ref. 40).
†Rate constants calculated by QM (B3LYP/6-31 + G(d)) at 298 K.

Figure 2: Key properties of six initial solvent and best solvent [1]

Algorithm

Find solvent properties using GC
And compute reaction rate constant k^{QM} of each solvent using DFT+SMD.

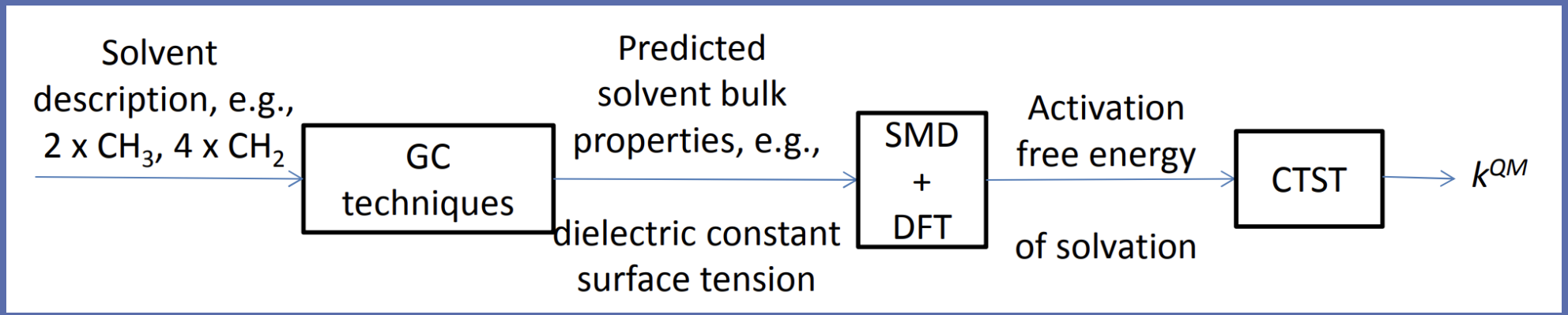


Figure 3: Step 2 of proposed algorithm [1]

Group Contribution (GC)

Add up the contribution of property from each functional group in a molecule. The functional groups that you have to consider is defined by you.

$$\boldsymbol{\Phi = f(\sum_{i \in G} n_i C_i)}$$

Φ = property

n_i = number of groups of type i in compound

C_i = contribution of group i to the property

i = group types

G = set of group types

Eqn. 1 Group contribution general equation[2]

What is SMD?

SMD is a Universal Solvation Model that calculate free solvation energy

Electrostatic Contribution

- **Treat solution as solute surround by dielectric medium (IEF-PCM)**
- **Defined electrostatic of solute using Poisson equation & Schrodinger equation**
- **Use Density Functional Theory to solve the equation[3][4]**

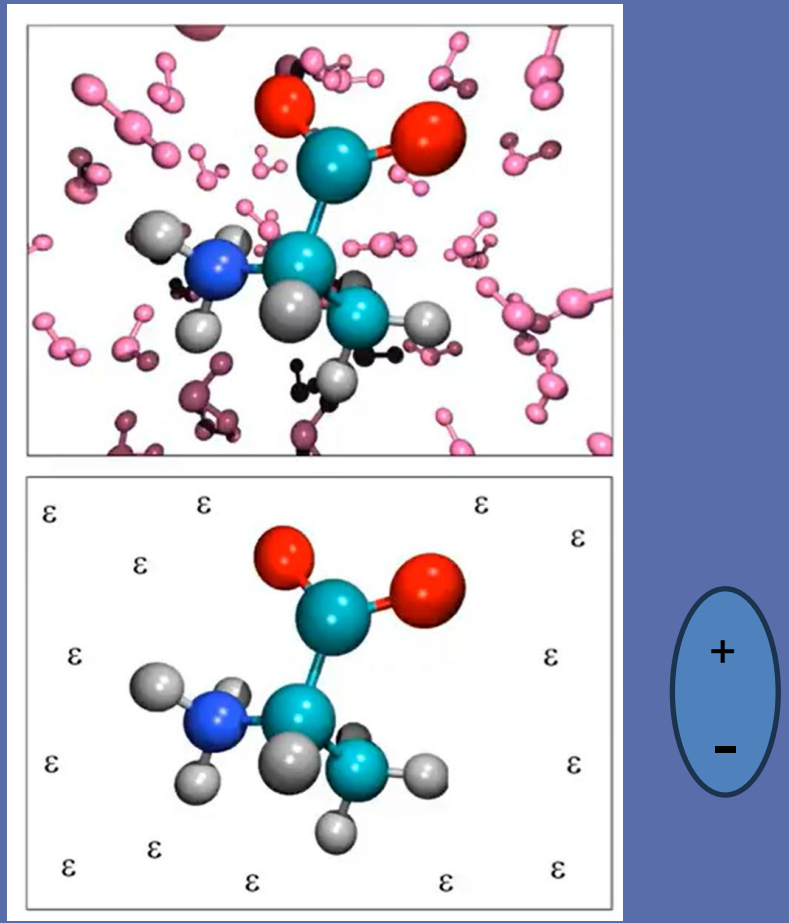


Figure 4: Illustration of IEF-PCM concept[3]

What is SMD?

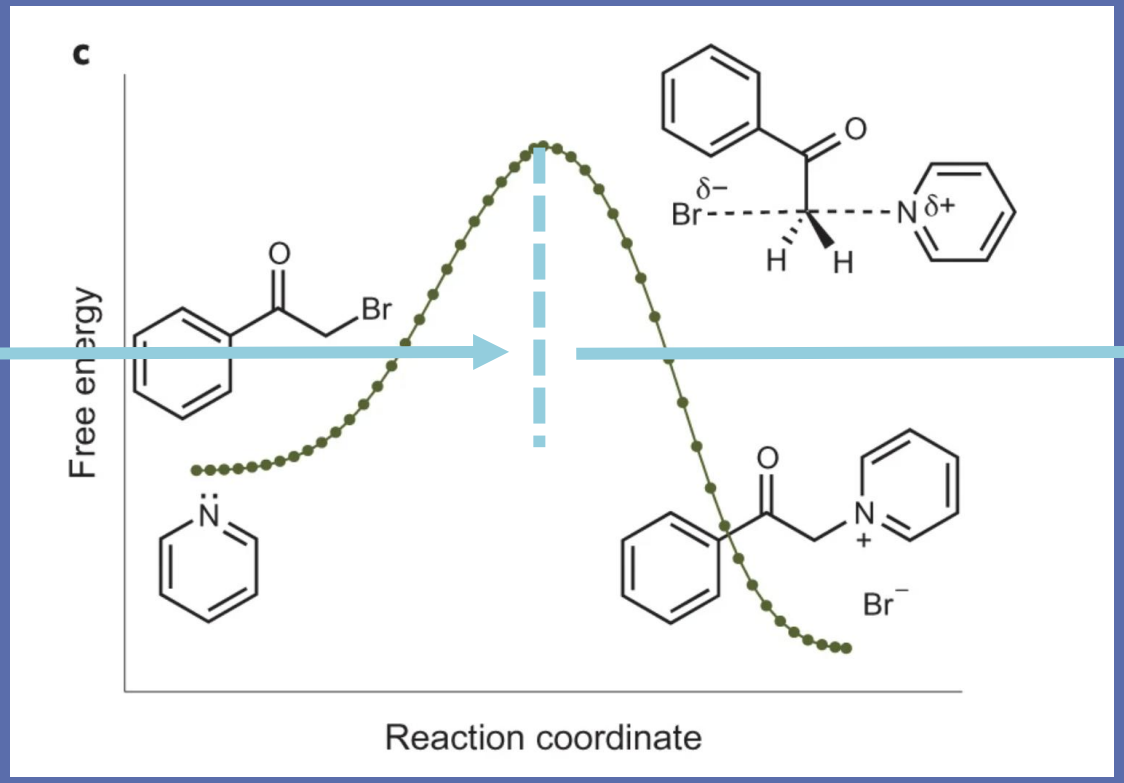
SMD is a Universal Solvation Model that calculate free solvation energy

- Sum of terms that are proportional to the solvent-accessible surface[4]



Conventional Transition-state Theory

Solvation Free Energy



$$k = Ae^{\frac{-E_a}{RT}}$$

Figure 5: Free energy curve [1]

Validity of SMD + QM-DFT Methodology

Compare SMD + QM DFT to ^{1}H NMR in situ kinetic experiments

- ❑ Predicted the trend but not the actual k value

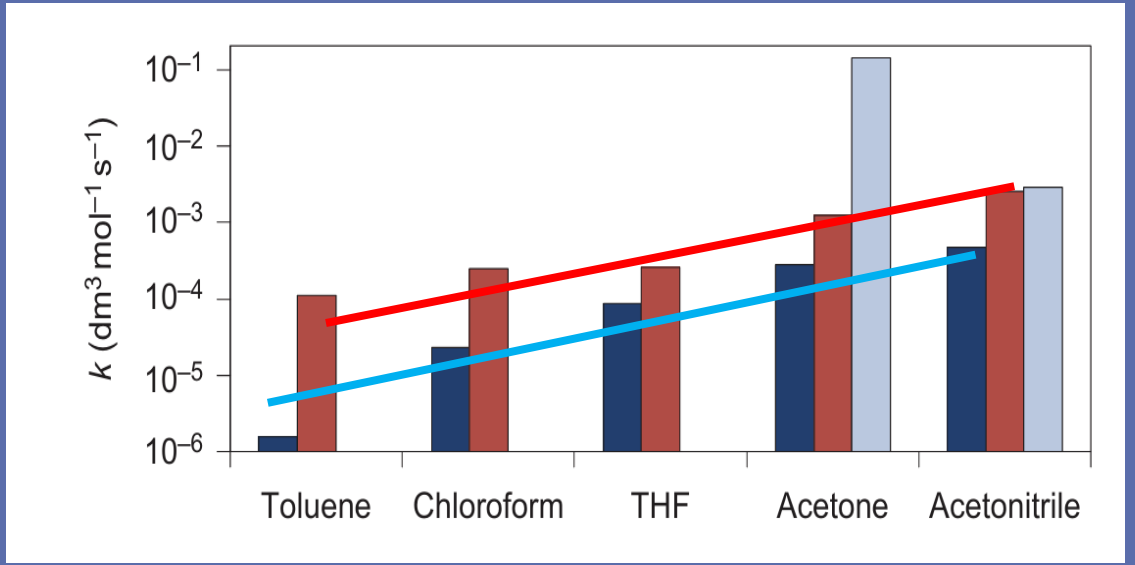


Figure 6. rate constant of different solvent estimate by different method. Dark blue = QM calculations, red = H NMR in situ experiments, light blue = conductance measurements[1]

Derived surrogate model from k^{QM}

Surrogate model: a model that approximates a more complex, computationally expensive model.[5]

Solvatochromic equation: related log k to C_i reaction specific coefficient

$$\log k = C_0 + C_A A + C_B B + C_S S + C_\delta \delta + C_H \delta_H^2$$

Eqn. 2 Solvatochromic Equation [1]

Table 2. All parameters known at this step

Symbol	Parameter	Absolute Error	Computational Method
--	Surface Tension	--	Group Characterization
--	Dielectric constant	--	Group Characterization
k^{QM}	QM rate constant	--	QM-DFT + SMD
A	Hydrogen bond acidity	0.017	
B	Hydrogen bond basicity	0.043	
S	Polarizability	0.065	Group Characterization
δ	Chemical class indicator	$1.13 MPa^{0.5}$	
δ_H^2	Cohesive energy density	$1.27 MPa^{0.5}$	

Mixed-Integer Linear Problem

Objective function: max k^{QM}

s.t $\Phi = f(\sum_{i \in G} n_i C_i)$

$$u_i = \begin{cases} 1 & \text{if the ith group in the molecule is the kth structural group in the basis set} \\ 0 & \text{Otherwise} \end{cases}$$ [6]

set of chemical feasibility and complexity constraints

$$log\, k = C_0 + C_A A + C_B B + C_S S + C_\delta \delta + C_H \delta_H^2$$

- Use a set of 38 functional groups, result in a design space of 1,341 possible combinations.
- Sensitive to initial solvent set.
- Isomeric solvents are not taken into account.

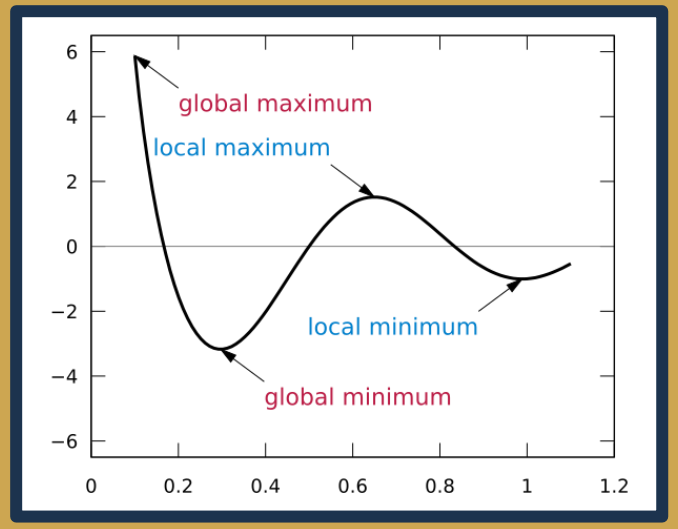


Figure 7. Local minimum & maximum [7]

Algorithm

Convergence criteria

If the candidate solvent from step 4 has not been identified previously

- The algorithm returns to step 2.
- The new reaction rate constant for the best solvent found in step 4 is added to step 2.

If the candidate solvent from step 4 has been identified previously

- The algorithm proceeds.

Algorithm

Experimental Verification

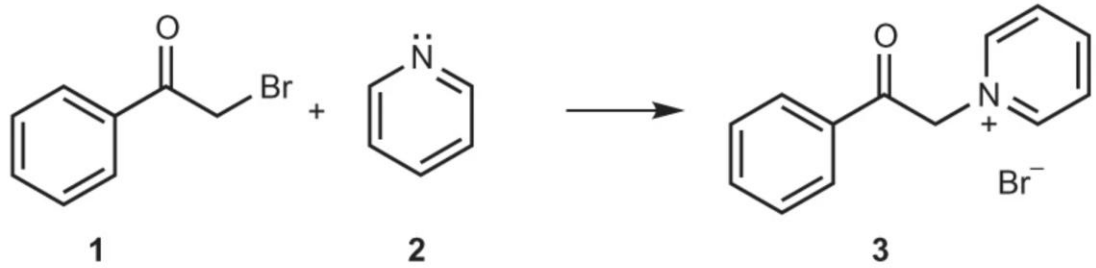
Evaluate the selected solvent against any additional criteria.

Confirm suitability through in situ kinetic experiments.



Main Results and Validation

❑ Application to a Menschutkin reaction



1 Phenacyl bromide

2 pyridine

3 Phenacyl pyridinium bromide

Initial Solvent Set:
Toluene
Chlorobenzene
Ethyle Acetate
THF
Acetone
Acetonitrile

Initial Constraints

Structure Basic Set

Algorithm

Iteration

Design Space of 1,341 combinations

Nitromethane

Main Results and Validation

❑ Application to a Menschutkin reaction

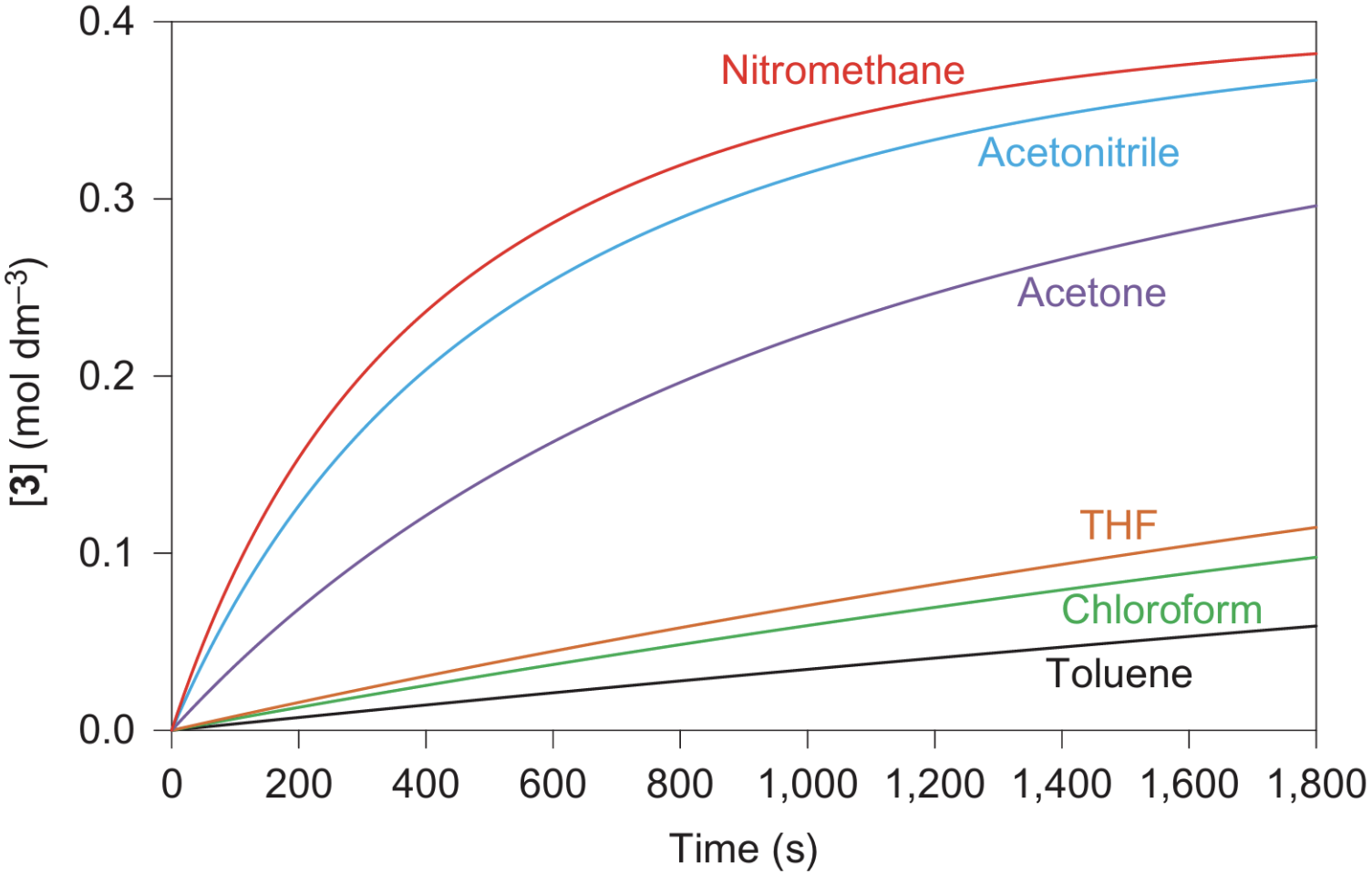


Figure 8: Concentration of product with different solvent [1]

- **Rarely prioritized solvent with 9 k^{QM} and 4 iterations**
- **Step 4 is solved in less than 1 CPU**
- **Nitromethane has a rate constant of 40% higher than initial best solvent**

Conclusions

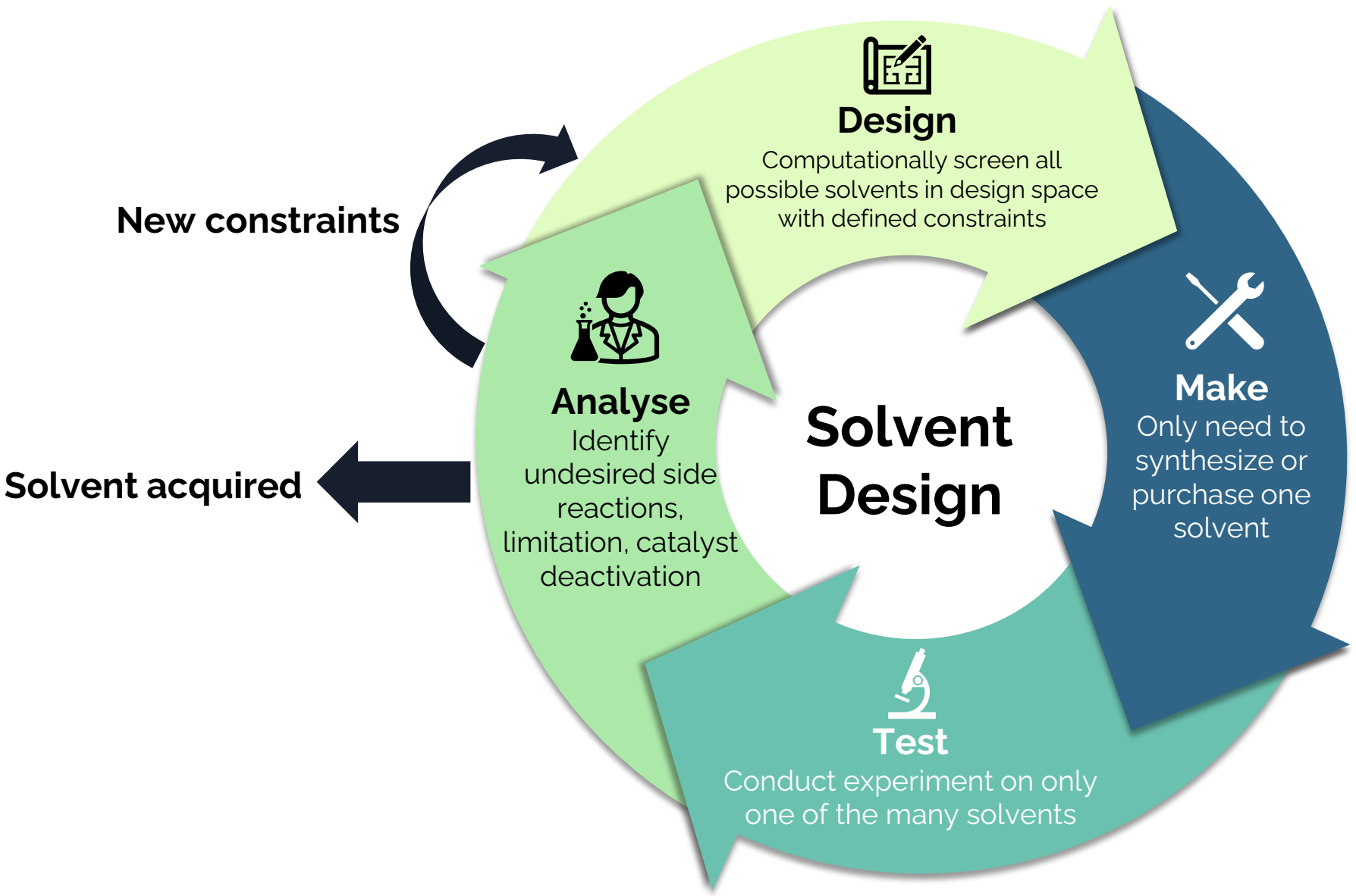
- Rapid identification of a solvent
- Computationally affordable
- Reduce amount of experiment work
- Allow quantitively evaluation of reaction rate in different solvent

Personal Highlight

Limitation

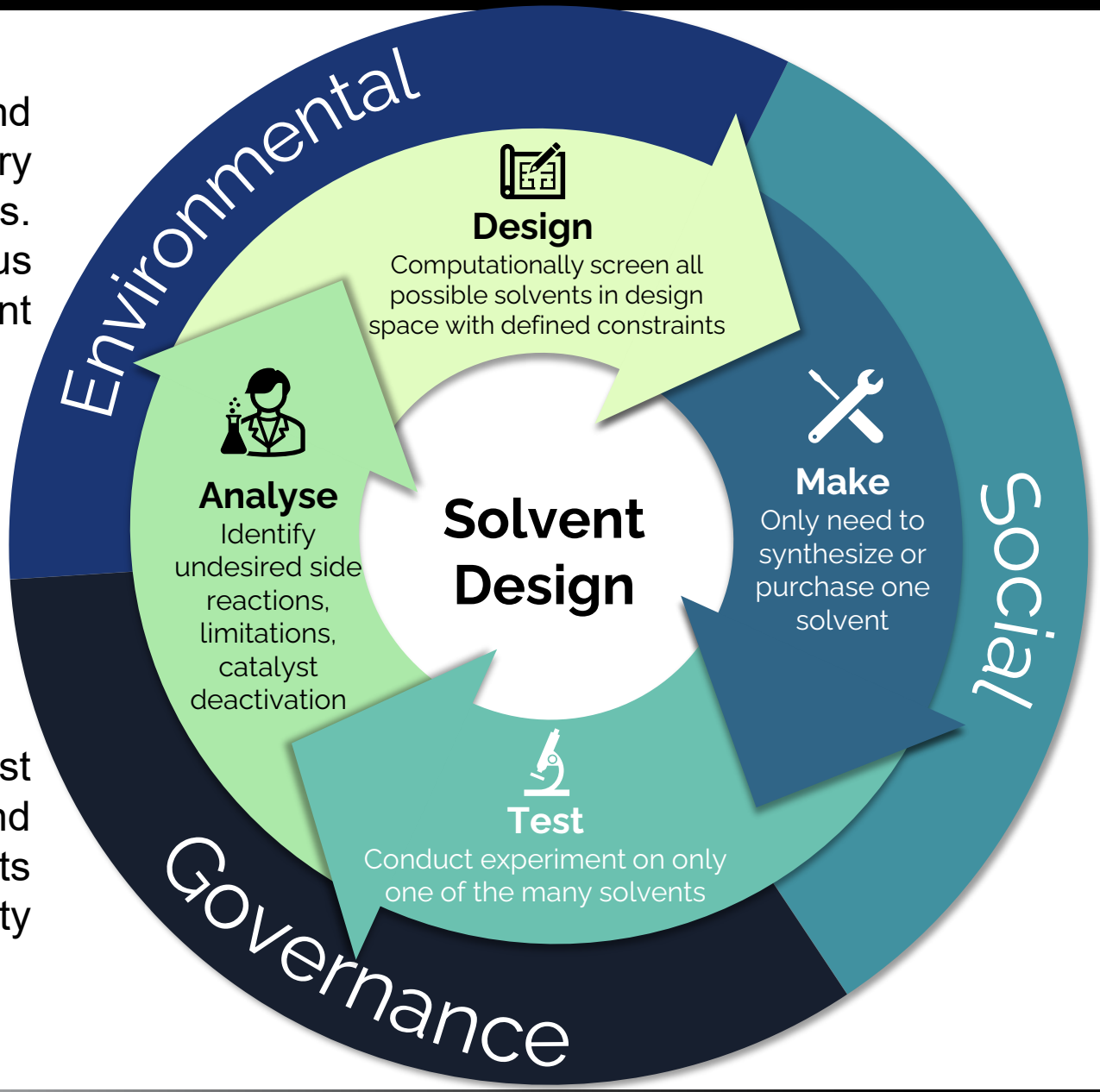
1. Have not been tried on many other reactions
2. Cannot distinguish isomeric solvents
3. Does not explain much about preventing catalyst deactivation
4. Regulation will be hard to approve as the SMD-DFT model is not fully developed and result is dependent on initial condition

Personal Highlight



Personal Highlight

Reduce consumption and waste in laboratory experiments.
Screen out hazardous solvent



Faster production of drug
Improve working environment of lab

Low computational cost
Cost Savings on Materials and Solvents
Increased Productivity

Reference

1. Struebing, H., Ganase, Z., Karamertzanis, P. et al. Computer-aided molecular design of solvents for accelerated reaction kinetics. Nature Chem 5, 952–957 (2013). https://doi.org/10.1038/nchem.1755
2. Lee, Lauren. *Design: Fundamentals of computer-aided molecular design (CAMD I,* UCL Lecture Note, 2024)
3. Chris Chramer. *CompChem.06.04 Solvation Models: Continuum (Implicit) Solvent Electrostatics*. Source: https://www.youtube.com/watch?v=tB4xZLUGJqM. [Cited On:11/11/2024]
4. Marenich, A. V., Cramer, C. J. & Truhlar, D. G. Universal solvation model based on solute electron density and on a continuum model of the solvent defined by the bulk dielectric constant and atomic surface tensions. *J. Phys. Chem. B* **113**, 6378–6396 (2009).
5. Williams, Bianca, and Selen Cremaschi. "Novel tool for selecting surrogate modeling techniques for surface approximation." Computer aided chemical engineering. Vol. 50. Elsevier, 2021. 451-456.
6. Poling, B. E., Prausnitz, J. M. & O'Connell, J. P. *The Properties of Gases and Liquids* (McGraw-Hill, 2000).
7. File:Extrema example original.svg. Wikimedia Commons. Source: https://commons.wikimedia.org/wiki/File:Extrema_example_original.svg. [Cited On:11/11/2024]
8. Liu, Si-Cong, et al. "DFT calculations in solution systems: solvation energy, dispersion energy and entropy." (*Physical Chemistry Chemical Physics* 25.2, 913-931 2023): 913-931.
9. Matt Timm, phD. Introduction to Density Functional Theory [Part One] Background. Source: https://www.youtube.com/watch?v=Ez_Fm4iTUeo . [Cited On:11/11/2024]

Supplementary Information

"SMD is a universal solvation model that treat observable solvation free energy as 2 main components

1. Electrostatic energy: arising from a self consistent reaction field treatment that involves the solution of the nonhomogeneous Poisson equation for electrostatics in terms of the integral equation formalism polarization continuum model (IEF-PCM)
2. Cavity-dispersion-solvent-structure : arising from short range interactions between the solute and solvent molecules in the first solvation shell. Sum of terms that are proportional to the solvent accessible surface areas of the individual atoms of the solute.

The SMD model employs a single set of parameters optimized over B3LYP/6-31 + G(d) which is an electronic structure method "

- Marenich, A. V., Cramer, C. J. & Truhlar, D. G. Universal solvation model based on solute electron density and on a continuum model of the solvent defined by the bulk dielectric constant and atomic surface tensions. *J. Phys. Chem. B* **113**, 6378–6396 (2009).

Supplementary Information Adopted from [3][9]

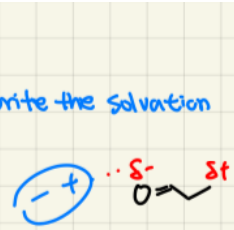
$\Delta G_{(ENP)} = \Delta G_{(EN)} + \Delta G_{(P)}$

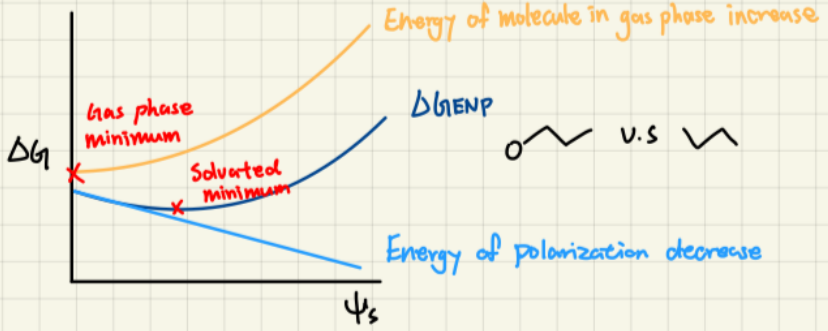
Gas phase

Dielectric constant = 1

Vaccuum

Polarization energy that favorite the solvation come from the interaction between solute and dielectric solvent





Define ΔG_{ENP} from Poisson Equation $G = -\frac{1}{2}\int_S \rho(s)\,\phi(s)\,ds$ (Charge on the surface)

Energy of Polarization

$$\Delta G_{ENP} = \left\{ H - \frac{1}{2}\left[\frac{2(\varepsilon-1)}{(2\varepsilon+1)}\right]\frac{\langle\psi|\mu|\psi\rangle}{a^3}\mu \right\} \psi = E\psi$$

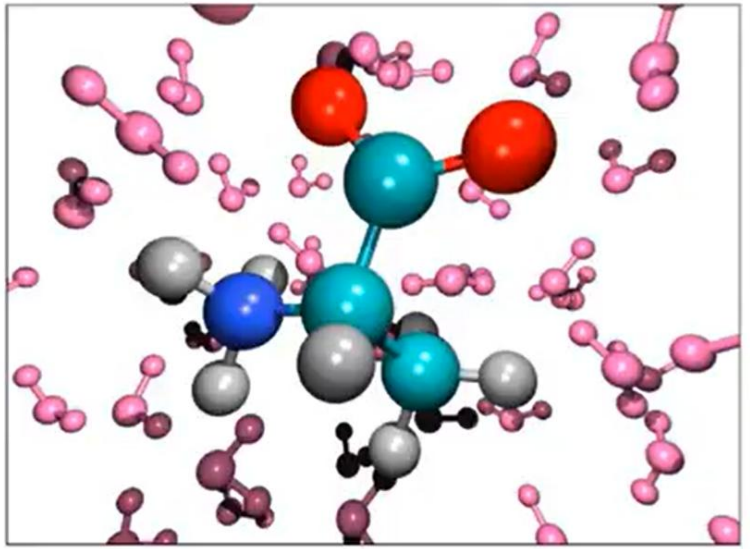
Hamilton Operator

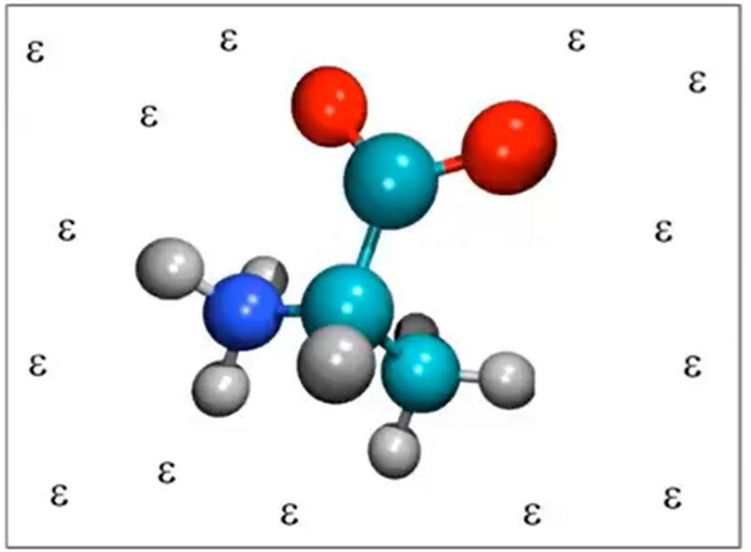
Energy in gas phase

Dipole moment as function of wavefunction

Radius of molecule

There are many ways to solve this and one is using density functional theory.





Supplementary Information Adopted from [3][9]

Density Functional Theory

Electron density

$$n(\vec{r}) = 2\sum_i \psi_i^*(\vec{r})\psi_i(\vec{r})$$

1. Ground state energy E from Schrödinger's equation is a unique functional of the electron density.

$$E[n(\vec{r})] = E$$

$$E[\{\psi_i\}] = E_{known} + E_{xc}$$

E_{known}: KE of electron; Columb interaction between electrons and nuclei; pair of nuclei; pair of electrons

E_{xc}: Other quantum mechanical effect

How do we find the $n(\vec{r})$ that minimize the total energy functional.

Solve set of equations (each involve only one electron)

$$\left[\frac{1}{2}\nabla^2 + V(\vec{r}) + V_H(\vec{r}) + V_{xc}(\vec{r})\right]\psi_i(\vec{r}) = \varepsilon_i\psi_i(\vec{r})$$

$\frac{1}{2}\nabla^2$: KE of e^-; $V(\vec{r}) + V_H(\vec{r}) + V_{xc}(\vec{r})$: Potential

$V(\vec{r})$ potential between e^- and nuclei

$V_H(\vec{r})$ Hartree potential, Columb repulsion and total electron density $V_H(\vec{r}) = \int \frac{n(\vec{r}')}{|\vec{r}-\vec{r}'|} d^3r'$

$V_{xc}(\vec{r})$ Correlation between exchange correlation contribution to the single electron equation

Supplementary Information (Adopted from [1])

Supplementary Table 2 – Initial molar concentrations of reactants in experiments and parameter estimation.

Experiment Number	$[\mathbf{1}]_w$ / mol dm^{-3}	$[\mathbf{1}]_{NMR}$ / mol dm^{-3}	$[\mathbf{1}]_{EST}$ / mol dm^{-3}	$[\mathbf{2}]_w$ / mol dm^{-3}	$[\mathbf{2}]_{NMR}$ / mol dm^{-3}	$[\mathbf{2}]_{EST}$ / mol dm^{-3}
1	0.10	0.11	0.11	0.17	0.19	0.20
2	0.18	0.18	0.18	0.35	0.35	0.37
3	0.27	0.26	0.26	0.48	0.47	0.51
4	0.36	0.35	0.35	0.48	0.46	0.52

Phenacyl bromide is denoted by [**1**], and pyridine by [**2**]. Experiments in nitromethane-d3, where the subscript w denotes the value obtained based on initial weighing; the subscript NMR denotes the value calibrated to the concentration of the internal standard; the subscript EST denotes the value obtained during the parameter estimation procedure.

Supplementary Information (Adopted from [1])

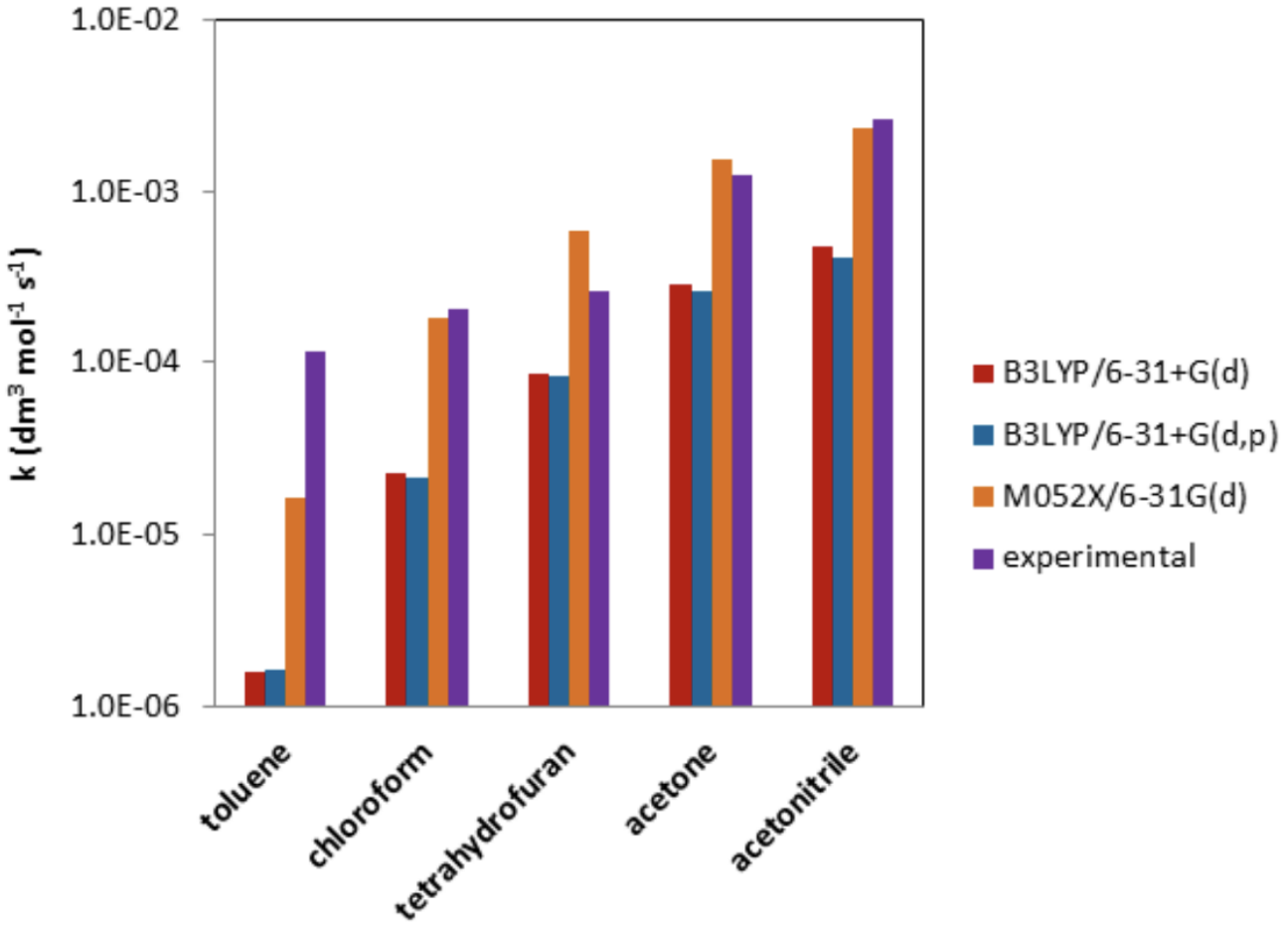
The 38 functional groups used in formulating the design problem are shown Supplementary Table 1.

Supplementary Table 1 – Functional groups used to build solvent molecules.

CH_3	CH_2	CH	C
CH=C	aCH	aC	$aCCH_3$
$aCCH_2$	aCCH	OH	aCOH
$CH_3C{=}O$	$CH_2C{=}O$	CH=O	CH_3COO
CH_2COO	CH_3-O	CH_2-O	CH-O
CH_2CN	COOH	CH_2Cl	CHCl
$CHCl_2$	aCCl	CH_2NO_2	$CHNO_2$
$CHNO_2$	I	aCF	CH_3CN
C_7H_5N	C_7H_8O	$CHCl_3$	$C_6H_5NO_2$
CH_3NO_2	C_4H_8O		

"aC" denotes a carbon in an aromatic ring.

Supplementary Information (Adopted from [1])



Supplementary Figure 5 - Comparison of rate constants calculated with 3 different computational models and experimental rate constants.

Supplementary Information (Adopted from [1])

Iteration 1

Step 2: Obtain reaction rate constants

Solvent	Rate constant [dm^3 mol^{-1} s^{-1}]
Toluene	1.61E-06
Chlorobenzene	3.94E-05
Ethyl acetate	4.88E-05
Tetrahydrofuran	8.70E-05
Acetone	2.86E-04
Acetonitrile	4.70E-04

Step 3: Build surrogate model

$\log k = -22.43 - 100.72\ A + 8.55\ B + 7.61\ S + 2.31\ \delta + 11.99\ \delta_H^2/100$

Step 4: CAMD solvent design

Best solvent found: Iodonitromethane

Groups	Number of groups
CH2NO2	1
I	1

k_{CAMD} (dm^3 mol^{-1} s^{-1}) 6.17E+11

Step 5: Test for convergence

New solvent generated - Continue

Iteration 2

Step 2: Obtain reaction rate constant for new solvent

Solvent	Rate constant [dm^3 s^{-1}]
Toluene	1.61E-06
Chlorobenzene	3.94E-05
Ethyl acetate	4.88E-05
Tetrahydrofuran	8.70E-05
Acetone	2.86E-04
Acetonitrile	4.70E-04
Iodonitromethane	**2.47E-04**

Step 3: Build surrogate model

$\log k = -1.32 - 13.40\ A - 7.61\ B + 3.36\ S - 4.13\ \delta - 1.20\ \delta_H^2/100$

Step 4: CAMD solvent design

Best solvent found: cf table; several isomers possible

Groups	Number of grou
CH2NO2	1
CH3	3
CH2	2
C=C	1

k_{CAMD} (dm^3 mol^{-1} s^{-1}) 2.22E-02

Step 5: Test for convergence

New solvent generated - Continue

Iteration 3

Step 2: Obtain reaction rate constant for new solvent

Solvent	Rate constant [dm^3 mol^{-1} s^{-1}]
Toluene	1.61E-06
Chlorobenzene	3.94E-05
Ethyl acetate	4.88E-05
Tetrahydrofuran	8.70E-05
Acetone	2.86E-04
Acetonitrile	4.70E-04
Iodonitromethane	2.47E-04
1xCH2NO2,3xCH3,2xCH2,1xC=C	**8.46E-05**

Step 3: Build surrogate model

$\log k = -4.50 - 0.84\ A - 1.11\ B - 0.11\ S - 1.46\ \delta + 1.21\ \delta_H^2/100$

Step 4: CAMD solvent design

Best solvent found: Nitromethanol

Groups	Number of groups
CH2NO2	1
OH	1

k_{CAMD} (dm^3 mol^{-1} s^{-1}) 8.25E-03

Step 5: Test for convergence

New solvent generated - Continue

Supplementary Information (Adopted from [1])

Step 4: CAMD solvent design

Best solvent found: Nitromethanol

Groups	Number of groups
CH2NO2	1
OH	1

k_{CAMD} (dm^3 mol^{-1} s^{-1}) 8.25E-03

Step 5: Test for convergence

New solvent generated - Continue

Iteration 4

Step 2: Obtain reaction rate constant for new solvent

Solvent	Rate constant [dm^3 mol^{-1} s^{-1}]
Toluene	1.61E-06
Chlorobenzene	3.94E-05
Ethyl acetate	4.88E-05
Tetrahydrofuran	8.70E-05
Acetone	2.86E-04
Acetonitrile	4.70E-04
Iodonitromethane	2.47E-04
1xCH2NO2,3xCH3,2xCH2,1xC=C	8.46E-05
Nitromethanol	**5.76E-04**

Step 3: Build surrogate model

$$\log k = -4.18 - 4.71\,A - 1.79\,B - 0.35\,S - 1.75\,\delta + 1.41\,\delta_H^2/100$$

Step 4: CAMD solvent design

Best solvent found: Nitromethane

Groups	Number of groups
CH3NO2	1

kCAMD 6.23E-02

Step 5: Test for convergence

New solvent generated - Continue

Iteration 5

Step 2: Obtain reaction rate constant for new solvent

Supplementary Information (Adopted from [1])

Iteration 5

Step 2: Obtain reaction rate constant for new solvent

Solvent	Rate constant [dm^3 mol^{-1} s^{-1}]
Toluene	1.61E-06
Chlorobenzene	3.94E-05
Ethyl acetate	4.88E-05
Tetrahydrofuran	8.70E-05
Acetone	2.86E-04
Acetonitrile	4.70E-04
Iodonitromethane	2.47E-04
1xCH2NO2,3xCH3,2xCH2,1xC=C	8.46E-05
Nitromethanol	5.76E-04
Nitromethane	**6.09E-04**

Step 3: Build surrogate model

$$\log k = -4.25 - 4.12\,A - 1.59\,B - 0.16\,S - 1.66\,\delta + 1.24\,\delta_H^2/100$$

Step 4: CAMD solvent design

Best solvent found: Nitromethane

Groups	Number of groups
CH3NO2	1
kCAMD	6.71E-04

Step 5: Test for convergence

No new solvent generated.

Step 6: Test best solvent found experimentally

Best solvent: Nitromethane